wants to be admitted to this university and he knows that he must score better than at least 70% of the students who took the test. Raghul takes the test and scores 585. Will he be admitted to this university?

6. The time taken to assemble a car in a certain plant is a random variable having a normal distribution of 20 hours and a standard deviation of 2 hours. What is the probability that a car can be assembled at this plant in a period of time .

   a) less than 19.5 hours?

   b) between 20 and 22 hours?

7. The annual salaries of employees in a large company are approximately normally distributed with a mean of \$50,000 and a standard deviation of \$20,000.

   (a) What percent of people earn less than \$40,000?

   (b) What percent of people earn between \$45,000 and \$65,000?

   (c) What percent of people earn more than \$70,00

8. X is a normally normally distributed variable with mean $\mu = 30$ and standard deviation $\sigma = 4$. Find

   (a) $P(x < 40)$

   (b) $P(x > 21)$

   (c) $P(30 < x < 35)$

9. The birth weight of babies is Normally distributed with mean 3,500g and standard deviation 500g. What is the probability that a baby is born that weighs less than 3,100g?

10. People's monthly electric bills in chennai are normally distributed with a mean of ₹ 225 and a standard deviation of ₹ 55. Those people spend a lot of time online. In a group of 500 customers, how many would we expect to have a bill that is ₹ 100 or less?

Summary

- Conditions for the binomial probability distribution are

  (i) the trials are independent

  (ii) the number of trials is finite

  (iii)each trial has only two possible outcomes called success and failure.

  (iv) the probability of success in each trial is a constant.

- The probability for exactly x success in n independent trials is given by

  $p(x) = \binom{n}{x} p^x q^{n-x}$ where $x = 0,1,2,3,...,n$ and $q = 1 - p$

- The parameters of the binomial distributions are n and p
- The mean of the binomial distribution is np and variance are npq
- Poisson distribution as limiting form of binomial distribution when n is large, p is small and np is finite.
- The Poisson probability distribution is $p(x) = e^{-\lambda} \frac{\lambda^x}{x!}$ $x = 0,1,2,3,\ldots$

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